

Total no. of Pages: 2

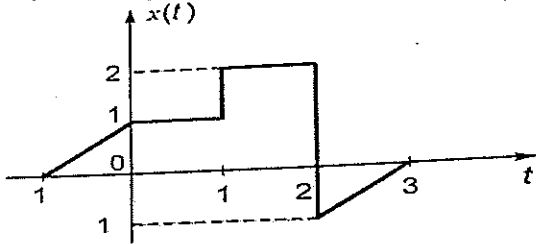
Roll no.

Degree:B.Tech...(EE)... Semester: IInd (Backlog)
MID-SEMESTER EXAMINATION, February, 2024

Course Title: Signal and System
Course Code: EEEEC03

Duration: 1:30 Hours

Max. Marks:25

Q. No.	Question	Marks	CO
1a	Find even and odd component of the signal (i) $x(t) = 1 + 2t + 3t^2$ (ii) $x(t) = \sin(t) \cdot u(t)$	3	CO-1
1b	Calculate the fundamental time period of the signal $x(t) = 1 + \sin\left(\frac{2\pi}{3}t\right) \cos\left(\frac{4\pi}{5}t\right)$	2	CO-1
2a	If a plot of signal $x(t)$ is as shown in the figure  then draw the plot of the signal $x(1-t)$.	2.5	CO-1
2b	Find the Convolution of two continuous time signals $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$	2.5	CO-1
3a	Find the response of discrete time LTI system having the following input and impulse responses. Find the Convolution of two continuous time signals $f[n] = a^n u[n]$ and $h[n] = a^n u[n]$	2.5	CO-1
3b	Discuss classification of signals	2.5	CO-1
4a	Find Laplace transform of the following signals and corresponding ROC. (i) $x(t) = e^{-t}u(t) + e^{-2t}u(t)$, (ii) $x(t) = e^{-t}u(t) + e^{2t}u(-t)$	2.5	CO-2
4b	Discuss Convolution Sum and Convolution Integral	2.5	CO-1

5a	Given that $X(s) = \frac{s^2+12}{s(s+2)(s+3)}$. Find the inverse Laplace of $X(s)$.	2.5	C0-2
5b	Discuss Laplace transform of elementary signals.	2.5	C0-2

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.